Q1) Prompting and inference-time methods

Qwen2.5-0.5B-Instruct (frozen weights, Colab T4, no paid APIs). Dev = GSM8K train 1–10, demonstrations = train 11–14, test = test 1–30 for every method. Test solutions were never shown to the model. Methods: zero-shot; four-shot (demonstrations as chat turns, greedy); self-consistency (four samples from the four-shot prompt, T=0.7, top-p=0.95, majority vote, ties to first-seen); reflection (greedy four-shot answer, then one revision call; 2 calls).

Development experiments

Only the dev set was used for decisions (one question = 10 points, so it served to find bugs, not to rank methods). Initial result (SYSTEM A, first parser): zero-shot 5/10, four-shot 3/10, self-consistency 6/10, reflection 6/10. Reading the wrong four-shot outputs showed one parser error (Weng: correct $10 in the text, but the fallback read the last number, 50), outputs ending in \boxed{} or a sentence, and genuine reasoning mistakes.

```
zero_shot          5/10 = 50.0% | calls=10 | generated_tokens=2537 | time=106s
four_shot          3/10 = 30.0% | calls=10 | generated_tokens=1537 | time=58s
self_consistency   6/10 = 60.0% | calls=10 | generated_tokens=7644 | time=97s
reflection         6/10 = 60.0% | calls=20 | generated_tokens=3465 | time=121s
```

- **Stricter prompt (SYSTEM B):** requiring the last line to be exactly #### <number>. Dev total fell from 20 to 13 correct (zero-shot 5→1, self-consistency 6→4, reflection 6→5, four-shot 3→3), so it was rejected.
- **Parser:** the final parser reads the last ####, then \boxed{}, then the last number. A last-line-only fallback misread \boxed{72} as 96 and was rejected. The final parser fixes \boxed cases but not Weng (no final line, ends with a sentence), so four-shot stays 3/10 on dev; I kept this as a known limitation rather than tune on 10 questions.

Settings were frozen before the final test run (SYSTEM A, final parser, greedy, max_new_tokens=512, seed 42). The test set was run more than once while debugging code, but no prompt or parser decision used test results.

Final test results (30 questions)

Method	Test accuracy	Total generated tokens	Model calls
zero_shot	11/30 (36.7%)	9,669	30
four_shot	11/30 (36.7%)	4,906	30
self_consistency	8/30 (26.7%)	29,800	120
reflection	11/30 (36.7%)	12,304	60

One question is 3.3 points, so differences up to 3 questions are within noise. Four-shot also reads 26,655 input tokens (demonstrations are resent each time).

Two cases

Helped (index 0, reflection). Janet's ducks (gold 18). Four-shot wrote 16−3−4 = 19 and answered 38. Reflection re-solved it step by step (3+4 = 7, 16−7 = 9, 9×$2 = $18) and was correct; it did not point out the slip, it produced a fresh, right solution.

Harmed (index 1, reflection and self-consistency). Robe with 2 bolts of blue and half as much white (gold 3). Four-shot was correct. Reflection invented a relation ("one blue needs 2 white") and

answered 1.5. Self-consistency got 5, 5, 3, 3 and the tie went to the first-seen wrong answer 5. Overall reflection helped 3 and harmed 3 questions (net 0); self-consistency helped 3 and harmed 6 (net −3).

Best trade-off

Zero-shot, four-shot and reflection all reach 11/30, so extra inference bought no accuracy: reflection costs 2× the calls and 2.5× the generated tokens, self-consistency 4× the calls and 6× the tokens and scores lower. Four-shot is the best trade-off in generated tokens and calls; zero-shot is an alternative if input tokens matter. A 0.5B model's errors are mostly systematic, so resampling or reviewing the same weak reasoning rarely helps.

Q2) Calculator-tool agent

Design. The agent writes `CALC: <expression>`, receives `OBSERVATION: <result or error>`, and ends with `FINAL: <number>` (max 3 calculator calls; four demonstrations built from train 11–14). The calculator parses with `ast.parse(mode=eval)` and evaluates only numeric constants, unary minus and + − * / // % ** through a whitelist; names, calls, attributes, strings and booleans are rejected, exponents above 100 are rejected, and errors return an ERROR string. `eval` and `exec` are never used.

Improvement. On dev (3/10) the model sometimes wrote an expression after FINAL (`FINAL: 48+24`) and the first code turned it into a wrong value (48 instead of 72). I made FINAL accept only a single number; otherwise the agent returns "FINAL must be a single number. Use CALC for the last step." Dev rose to 4/10. This message never appeared in the 30 test traces, so the test result is unaffected.

Method	Correct (/30)	Input tokens	Generated tokens	Generations
Four-shot (Q1)	11	26,655	4,906	30
Calculator agent	8	55,201	756	68 (38 calc calls)

The agent was less accurate and read about twice the input tokens (demonstrations and transcript resent each turn), but generated far fewer tokens. The bottleneck of the small model is translating the story into an expression, not arithmetic.

Failure 1, formulation (index 18). "How many dozens of eggs will she eat in 4 weeks?" (3 eggs a day; gold 7 = 84/12). The model wrote CALC: 3*7*24*4 = 2016, treating "dozens" as a factor of 24 instead of dividing by 12. The calculator was right; the expression was wrong.

Failure 2, formulation and unused observation (index 25). Marie paid $50; how many $8.50 pizza boxes (gold 2)? The model wrote CALC: 12+3*5+4*1.5+8.5 = 41.5, adding one box price instead of solving for the unknown count, then answered FINAL: 50, the total from the question, not the observation. Arithmetic was correct in both cases; index 2 shows a related protocol failure ($ and commas gave syntax errors, then an unsupported FINAL).

Why successful execution is not correctness. The calculator only guarantees that an expression is evaluated correctly. It cannot know whether the expression models the question, or whether the model used the returned value. In index 18 the tool succeeded and the answer was wrong; in index 25 it succeeded and was ignored. Correctness needs an independent check (a second formulation, sanity bounds, or a known answer).

Q3) Critique of Ferber et al. (MIRA)

Flaw: physical examination is reduced to retrieving documented text. In the evaluation environment the examination is not performed or observed. MIRA and the physicians call a tool that returns findings already written in the MIMIC-IV record: “PhysicalExaminationRequest is used to document and retrieve results from physical exams” (Methods, tool list). The patient agent is only a text model that “does not have access to external tools” and answers from the history of present illness (Methods, ‘Patient agent definition’); it cannot show appearance, gait, breathing effort or other non-verbal findings.

Where it matters. The paper compares examination behaviour as a request rate: “MIRA requested physical examinations more consistently than human physicians (on average 97.1% versus 87.8% … and 88.4% …)” (Results, Fig. 3c). This metric only counts whether the examination was requested. It does not test whether the examination was done well or its findings interpreted, and a physician who knows the findings are a retrieved record may reasonably skip or abbreviate the request. The comparison therefore measures interface behaviour more than clinical examination skill.

Effect. Physician performance may be understated and agent performance overstated, because the examination channel physicians rely on (observation, palpation, auscultation, non-verbal cues) is replaced by a lookup that suits a text-native agent. Findings are only those documented at admission, so the agent is never tested on cases where the examination would change the diagnosis. Claims about autonomous clinical performance are therefore not fully supported by this setting.

Related limit: the simulated patient. The authors report high faithfulness (Fig. 2a: 99.4% consistency between original and rephrased answers, 99.3% alignment with the HPI) and instruct the agent to say it does not know when information is missing. These checks used 622 sampled question–answer pairs with a physician rater and an LLM judge, so rare invented or inconsistent details cannot be excluded, and they were not measured per diagnosis path. The authors also concede that the responses “may yield responses that are more structured than real speech of patients” (Discussion, limitations); the paper argues this affects all groups equally.

Fixes. Add examination steps that require observation (e.g. images or video, standardised patients, or an examiner who reports findings only when the correct manoeuvre is requested); report examination quality and its effect on the diagnosis, not only request rate; analyse cases where the examination is decisive separately; audit the simulated patient with a larger human review and an explicit “not documented” category; and frame the conclusion as performance in a text-and-record setting.

Github Repo:

https://github.com/bilgili20/Agentic_AI_assignment/tree/main